18 (a) Show that $(2x + 1)(x + 3)(3x + 7)$ can be written in the form $ax^3 + bx^2 + cx + d$ where a, b, c and d are integers.

$(2x + 1)(x + 3) = 2x^2 + 7x + 3$

1 mark
Expand first two brackets.

Multiply by $(3x + 7)$:

$6x^3 + 14x^2 + 21x^2 + 49x + 9x + 21$

1 mark
Multiply all six terms.

$= 6x^3 + 35x^2 + 58x + 21$

1 mark
Collect correct cubic.

(3)

(b) Solve $(1 - x)^2 < \frac{9}{25}$

$|1 - x| < 3/5$

1 mark
Take both square-root signs.

$-3/5 < 1 - x < 3/5$

1 mark
Solve both critical values.

$2/5 < x < 8/5$

1 mark
Open interval 2/5 to 8/5.

$2/5 < x < 8/5$

(3)

(Total for Question 18 is 6 marks)